Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of full-stack SaaS development in **property management and leasing** - the same domain, same customer type, same scale. Shipped 20+ production modules at Entrata (100K+ MAU, 1M+ property records, 120+ Agile sprints) covering leasing workflows, pricing calculations, and affordable-housing compliance automation. Built AgentProbe — an open-source agentic AI toolkit actively used to accelerate development workflows — and LeaseI, a production-grade agentic pipeline for lease document analysis. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Built and owned **leasing and property management workflows at scale** — 20+ production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records; 99.5% uptime over 6+ consecutive months.
- Improved **algorithm and database performance** — tuned 200+ SQL queries for 30% performance gain across all client environments; designed schema and data access layers for 20+ production modules supporting complex leasing and pricing calculations.
- Developed **affordable housing compliance modules** (LIHTC/Tax Credit, Section 8/HCV, HUD) — automated income eligibility verification, AMI threshold calculations, and recertification workflows across 100K+ unit portfolios; eliminated 80% of manual operational workload.
- Cut microservices execution time 50% via Docker containerization; achieved **100% PHI compliance** on AWS S3 migration (zero security incidents); mentored 5+ engineers through 300+ peer code reviews.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%.

PROJECTS

AgentProbe: Multi-Agent AI Development & Testing Toolkit

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built an agentic AI toolkit that accelerates development workflows — 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, multi-provider LLM routing, and a React dashboard tracking regression drift and latency; deployed as a 15-endpoint FastAPI backend with CI/CD quality gates.
- Implemented a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with automated quality gates — demonstrates the agentic coding practices and eval-first development culture this team is building toward.

LeaseI: Agentic Pipeline for Lease Document Analysis

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an agentic LangGraph pipeline that autonomously audits 1,240 residential lease PDFs for compliance gaps across 10 rules, 5 US states + federal standards — **93.6% precision, 82.9% recall, 87.9% F1**; deterministic rules engine keeps decisions fully auditable; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with regression gate.

CrashClarity: Large-Scale Algorithm Development (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ records — Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables; deployed interactive analytics dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

AI & Agentic Coding: LangGraph, LangChain, Agentic AI, Multi-Agent Systems, RAG Pipelines, LLM-as-Judge, LLM Evaluation, Prompt Engineering, ChromaDB, OpenAI API, Anthropic API, FastAPI, REST APIs, GitHub Copilot, Cursor

Full-Stack & Data: React, TypeScript, PostgreSQL, MySQL, Schema Design, SQL Optimization, NumPy, Pandas, Scikit-learn, PyTorch, Feature Engineering, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Microservices, ETL, Data Pipelines, RabbitMQ, Git, Linux, Agile/Scrum (120+ sprints)